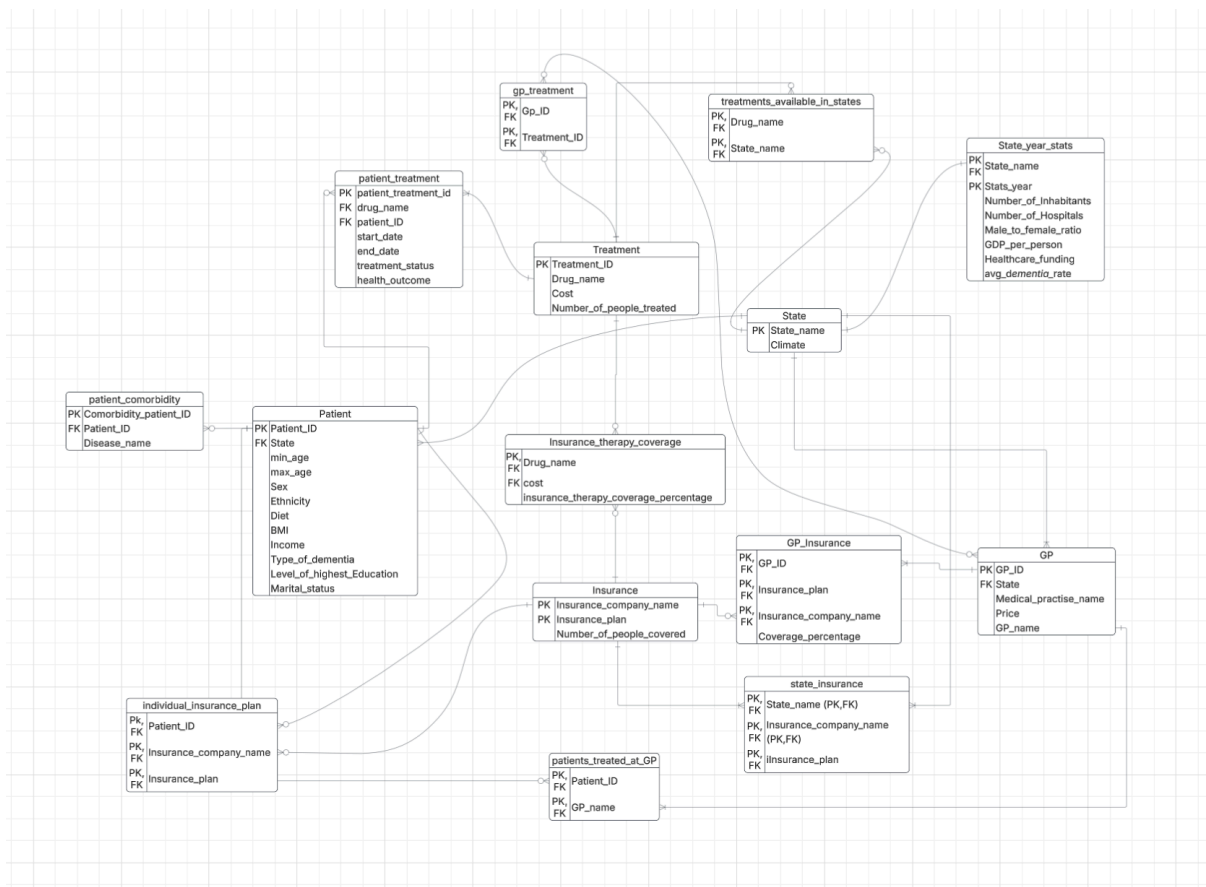


Final ERD before Normalization



Final Normalization

First Normal Form

With respect to the first phase of the project we added two new relations, “patient_comorbidity” and the independent entity “State_year_stats”. These two were constructed in order to purposely avoid first normal form.

The relation “patient_comorbidity” was in fact created to handle 1NF inside the patient table, by extracting each comorbidity to allow patients to have multiple values without violating 1NF.

Also, while refining patient_comorbidity for 1NF, we identified that the surrogate key (Comorbidity_patient_ID) alone did not prevent the same comorbidity from being recorded twice for the same patient, since Comorbidity_patient_ID is always unique regardless of the actual (Patient_ID, Disease_name) pair. To address this, we added a UNIQUE (Patient_ID, Disease_name) constraint, ensuring each comorbidity is stored only once per patient while preserving the surrogate key for simple referencing from other tables.

As for “State_year_stats” instead, it is composed of many previous attributes belonging to the relation “State”, to allow tracking annual state metrics over time rather than storing a single static snapshot.

In conclusion, we found no further violation of 1NF, neither a table containing multiple values, nor any repeating group of attributes.

Second Normal Form

For a table to be in 2NF it is required that it satisfies 1NF and that there is no partial dependency on just one of the attributes of a composite key.

Hence, every 1NF table that does not have a composite key is automatically 2NF, thus we evaluate all the relations with a composite key. These are: “State_year_stats”, “Insurance”, “GP_insurance”.

It is worth noticing that in “GP_insurance” the attribute “coverage_percentage” does not have any partial dependency. This is because the attribute naturally depends on “GP_ID” (it is in fact the percentage of coverage for a GP visit by insurance) but also both on the “Insurance_plan” and “Insurance_company_name”:

The last two attributes were constructed as a composite key also in the relation “Insurance”. This was done in order to prevent confusion, since many companies can share a similar plan name, and it would be impossible to uniquely identify the table without them. This is why 2NF holds also for “Insurance_company_name” since being a bridge table it covers the “Insurance” composite key as well.

A similar reasoning can be applied to the “State_year_stats” relation, where the composite is there for the purpose of uniquely identifying the table, and in doing so it creates no partial dependency.

Overall, our conclusion is that the relational schema does not violate 2NF.

Third Normal Form

Third Normal Form (3NF) requires the database to be in 2NF and that no transitive dependencies exist among non-key attributes ($X \rightarrow Y$ where X is not a superkey).

First of all, we found the “Insurance_therapy_coverage” table to be redundant and transitively dependent on “drug_name”. This is because cost is a foreign key referencing “Treatment(cost)”, however in “Treatment” the cost is already determined by the “drug_name”.

$$\text{drug_name} \rightarrow \text{Treatment.cost} \rightarrow \text{cost}$$

To avoid this, we removed cost from the “Insurance_therapy_coverage” table and retrieved it with a join query that can be found in “Additionalqueries.SQL” in the “SQL Code” section of the repository.

As for the “GP” table, The table contains GP_ID, Medical_practise_name, and State. If multiple GPs work at the same medical practice, and that practice is located in a specific State, a transitive dependency exists:

$GP_ID \rightarrow Medical_practise_name \rightarrow State$

In such a case, a convenient solution would be creating a new relation for “Medical_practise” with “Medical_practise_name” as a primary key and “State” as a foreign key.

Final ERD

